Prokaryotic Genome Assembly and Annotation Result Summary

Sample ID: AH3_raw

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Important note regarding the output files:

This document contains a brief summary of our findings for **Sample ID:** AH3_raw. The major output files needed for publication of these findings are given in the "1.major_output_files" folder. There are 6 other folders (2.assembled_genome, 3.annotated_genome, 4.metabolic_pathways, 5.antibiotic_resistant_genes, 6.biosynthetic_gene_clusters, 7.crisper_cas_sites) provided. These folders contain all the files generated during each particular analysis.

In the "1.major_output_files" folder there are following files:

File Name	Description
quality_control_summary.html	Summary of quality control (trimming) in .html format. Open the file using any web browser.
assembled_genome_final.fasta	Final assembled genome in .fasta format. Open the file using notepad in windows or any text editor in linux.
annotated_genome.fna	Annotated genome in .fna format. Open the file using notepad in windows or any text editor in linux.
annotated_genome.fsa	Annotated genome in .fsa format. Open the file using notepad in windows or any text editor in linux.
annotated_genome.gbk	Annotated genome in .gbk format. Open the file using notepad in windows or any text editor in linux.
annotated_genome.gff	Annotated genome in .gff format. Open the file

	using notepad in windows or any text editor in linux.
annotation_summary.txt	Annotation summary in .txt format. Open the file using notepad in windows or any text editor in linux.
annotation_table.tbl	Annotation table in .tbl format. Open the file using notepad in windows or any text editor in linux.
annotation_table_main.tsv	Main Annotation table in .tsv format. Open the file using microsoft excel in windows or libre office calc in linux.
genes.ffn	Identified genes and their sequences in .ffn format. Open the file using notepad in windows or any text editor in linux.
proteins.faa	Identified proteins and their sequences in .faa format. Open the file using notepad in windows or any text editor in linux.
metabolic_summary__module_completeness.tab	Summary of metabolic pathway modules and the percentage of their completeness in .tab format. Column 2 contains the specific pathway module name. Column 3 contains the pathways category name in a broader sense. Column 4 contains how much (%) of the pathway is covered by the genes in your genome e.g. 100 means 100% of the genes involved in that pathway module is present in the sample. Open the file using microsoft excel in windows or libre office calc in linux.
metabolic_summary__heatmap.pdf	Heatmap of metabolic pathways in .pdf format. It contains a graphical representation of the pathway modules and their completeness (%). Open the file using an adobe acrobat reader or any web browser.
antibiotic_resistant_genes.tsv	List of detected antibiotic resistant genes in .tsv format. Column 5 (GENE column) contains the names of antibiotic resistant genes. Open the file using microsoft excel in windows or libre office calc in linux.
biosynthetic_gene_clusters.html	List of identified biosynthetic gene clusters (BGCs) that encode secondary metabolites in .html format. Open the file using any web browser.
crisprs_all.tab	List of CRISPR repeat sequences and

	corresponding Cas protein subtypes in .tab format. Open the file using microsoft excel in windows or libre office calc in linux. If you do not see this file in the "1.major_output_files" folder, it means no CRISPR site has been detected in this sample.
crispr_spacers.fa	List of CRISPR spacer sequences .fa format. If you do not see this file in the "1.major_output_files" folder, it means no CRISPR spacer has been detected in this sample.
crispr_cas_plot.png	Graphical view of CRISPR Cas system in .png format. If you do not see this file in the "1.major_output_files" folder, it means no CRISPR Cas system has been detected in this sample.

Methodology:

Step	Software Name	Reference
Trimming	Fastp	(1)
Genome assembly	Unicycler	(2)
Genome annotation	Prokka	(3)
Metabolic pathway detection	MicrobeAnnotator	(4)
Biosynthetic gene cluster identification	AntiSMASH	(5)
Antibiotic resistance gene identification	Abricate	(6)
CRISPR Cas identification	CRISPRCasTyper	(7)

Genome Assembly and Annotation summary:

No. of contigs: 33

Detected Organism: *Aeromonas diversa strain*

No. of Bases: 4105655

No. of genes: 3967

No. of mRNA: 3967

No. of miscellaneous RNA: 63

No. of rRNA: 3

No. of tRNA: 88

No. of tmRNA: 1

Assembled_genome_final.fasta file in the "1.major_output_files" folder is the final assembled genome file. Annotation files in various formats are available in the "3.annotated_genome" folder. A copy of the most commonly used annotation files have also been kept in the "1.major_output_files" folder.

Summary of Major Metabolic Pathways:

8 Metabolic pathway modules have been detected with 100% completeness.

Check the metabolic_summary__module_completeness.tab file in the "1.major_output_files" folder to see the names and completeness (%) values of the metabolic pathway modules.

Summary of Identified Biosynthetic Gene Clusters:

0 biosynthetic gene clusters were identified.

Summary of Identified Antibiotic Resistant Genes:

3 antibiotic resistant gene was identified as follows:

ampH_1

sul2_2

tet(A)_4

Check the antibiotic_resistant_genes.tsv file in the "1.major_output_files" folder to get more details about the identified antibiotic resistant genes.

Summary of Identified CRISPR sites:

1 CRISPRs spacer was identified.

Check the crisprs_all.tab and the crispr_cas_plot.png files in the "1.major_output_files" folder to get more details about the identified CRISPR Cas systems.

References:

1) Chen, S., Zhou, Y., Chen, Y., & Gu, J. (2018). fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics (Oxford, England)*, *34*(17), i884–i890. https://doi.org/10.1093/bioinformatics/bty560
2) Wick, R. R., Judd, L. M., Gorrie, C. L., & Holt, K. E. (2017). Unicycler: Resolving bacterial genome assemblies from short and long sequencing reads. PLoS computational biology, 13(6), e1005595. https://doi.org/10.1371/journal.pcbi.1005595
3) Seemann T. (2014). Prokka: rapid prokaryotic genome annotation. *Bioinformatics (Oxford, England)*, *30*(14), 2068–2069. https://doi.org/10.1093/bioinformatics/btu153
4) Ruiz-Perez, C. A., Conrad, R. E., & Konstantinidis, K. T. (2021). MicrobeAnnotator: a user-friendly, comprehensive functional annotation pipeline for microbial genomes. *BMC bioinformatics*, *22*(1), 11. https://doi.org/10.1186/s12859-020-03940-5
5) Medema, M. H., Blin, K., Cimermancic, P., de Jager, V., Zakrzewski, P., Fischbach, M. A., Weber, T., Takano, E., & Breitling, R. (2011). antiSMASH: rapid identification, annotation and analysis of secondary metabolite biosynthesis gene clusters in bacterial and fungal genome sequences. *Nucleic acids research*, *39*(Web Server issue), W339–W346. https://doi.org/10.1093/nar/gkr466
6) https://github.com/tseemann/abricate
7) Russel, J., Pinilla-Redondo, R., Mayo-Muñoz, D., Shah, S., & Sørensen, S. (2020). CRISPRCasTyper: Automated Identification, Annotation, and Classification of CRISPR-Cas Loci. The CRISPR Journal, 3(6), 462-469. doi: 10.1089/crispr.2020.0059